

Markscheme

Mathematics: analysis and approaches

Practice question bank

61. (a)

The sum of the first n terms of a sequence is given by $S_n = n^2 + 4n$ for all positive integers n .

Find an expression for the general term u_n , and hence find the value of u_{15} .

$$\text{use } u_n = S_n - S_{n-1}: u_n = (n^2 + 4n) - ((n-1)^2 + 4(n-1)) \quad (M1)$$

$$u_n = 2n + 3 \quad \mathbf{A1}$$

$$\text{substitute } n = 15: u_{15} = 2(15) + 3 \quad (M1)$$

$$u_{15} = 33 \quad \mathbf{A1}$$

[5 marks]

(b)

Hence find the value of $S_{10} - S_9$.

$$S_{10} - S_9 = u_{10} = 2(10) + 3 \quad (M1)$$

$$= 23 \quad \mathbf{A1}$$

[2 marks]